LESSON 4.3

Estimate the Percent of a Number

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can estimate the percent of a number using benchmark percents, rounding, and compatible numbers.

WHY THIS MATTERS Margaret checked the cloud storage on her phone: 6 GB total, 4 GB used. Approximately what percent of her storage is used — and of the used storage, approximately what percent are photos?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
estimate *Spanish: estimación*	A reasoned answer that is close enough to be useful, found without exact computation.	about 70% used
benchmark percent *Spanish: porcentaje de referencia*	A friendly percent — 1%, 10%, 25%, 50%, 75%, 100% — used as a landmark for estimating.	10%, 25%, 50%, 75%
compatible numbers *Spanish: números compatibles*	Numbers close to the originals that are easy to compute with mentally.	8.6% of 216 → 10% of 220: both nudged to numbers that divide nicely.
double number line *Spanish: recta numérica doble*	Two parallel number lines that pair amounts with their percents at the same positions.	0, 30, 60 GB over 0%, 50%
the whole *Spanish: el todo*	The quantity that counts as 100% in a problem — and it can change between questions.	all of the storage
Decompose *Spanish: Descomponer*	Break a percent into friendly pieces that are easy to find.	15% = 10% + 5%

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A game download is 80 MB. 41 MB are done. About what percent is done?

1. Whole: 80 MB is 100%.
2. 50% of 80 is 40 MB.
3. 41 MB is just over 40 MB.

Answer: about 50%

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

The Green-Team recycled 216 pounds of material; 8.6% was metal.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The whole in this problem is _____ , so it stands for one hundred percent.
- The benchmark closest to _____ is _____ .
- My estimate is about _____ % because _____ .
- I know my estimate is reasonable because _____ .

Word bank estimate | about | benchmark | whole | part | double number line | half | one quarter | three quarters | close to | reasonable | compare

WATCH OUT!

A common mistake

Keeping the same whole for every question

LESSON 4.3

Estimate the Percent of a Number

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Margaret checked the cloud storage on her phone: 6 GB total, 4 GB used. Approximately what percent of her storage is used — and of the used storage, approximately what percent are photos?

PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

Margaret's phone has 6 GB of storage and 4 GB is used. About what percent is used?

(A) About 25%
(B) About 40%
(C) About 70%
(D) About 150%

HINT What is the whole?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

On the double number line for 6 GB, what amount lines up with 25%?

(A) 1.25 GB
(B) 1.5 GB
(C) 2.5 GB
(D) 4 GB

HINT 25% means one quarter.



0% 50% 100%

WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

3 **Circle the letter of the best answer. Show how you know.**

Which benchmark percents make good landmarks on a number line?

(A) 0%, 25%, 50%, 75%, 100%
(B) 3%, 17%, 62%, 94%
(C) Only 50%
(D) Only 100%

HINT Which percents can you find mentally?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

To estimate 8.6% of 216 by rounding, which numbers should you use?

(A) 1% of 216
(B) 8% of 300
(C) 10% of 220
(D) 86% of 216

HINT Round each number to something friendly.

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

What is 10% of 220?

(A) 2.2
(B) 22
(C) 44
(D) 220

HINT Write 10% as a decimal.



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

PART 2 Explain and correct

reasoning

6 Answer in complete sentences.

The word 'approximately' appears in the storage problem. What does it tell you about the answer you need?

SENTENCE STARTER 'Approximately' tells me ___ .
So instead of computing exactly, I can ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

You sorted a move into 'Benchmark percent' and your partner into 'Compatible numbers.' Which number did each strategy change — the percent or the whole — and why does changing that one keep the estimate close?

SENTENCE STARTER This is ___ because it changes the ___, and the estimate stays close since ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help

LESSON 4.3

Estimate the Percent of a Number

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Margaret checked the cloud storage on her phone: 6 GB total, 4 GB used. Approximately what percent of her storage is used — and of the used storage, approximately what percent are photos?

PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

The Green-Team recycled 216 pounds; 8.6% was metal. Using compatible numbers 10% and 200, the estimate is:

(A) 2 pounds
(B) 20 pounds
(C) 86 pounds
(D) 208 pounds



0% 50% 100%

WORK

2 **Circle the letter of the best answer. Show how you know.**

Photos use 3.5 GB of the 4 GB of used storage. About what percent of the USED storage is photos?

(A) About 35%
(B) About 58%
(C) About 87%
(D) About 99%

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Margaret has a 6 GB phone with 4 GB used. Question A asks what percent of the total 6 GB is used. Question B asks what percent of the used 4 GB is photos. Why is the whole different in the two questions?

(A) The phone gained storage between the questions

(B) The whole never changes in a percent problem

(C) The second question asks about the used storage, so 4 GB becomes 100%

(D) Because photos are more important than mail

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Estimate 24% of 81 with benchmarks.

(A) About 2

(B) About 8

(C) About 20

(D) About 40

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Your estimate for 8.6% of 216 was 22 pounds (using 10% of 220). Is the exact answer higher or lower than 22?

(A) Higher, because rounding always shrinks the answer

(B) Lower, because both 10% and 220 rounded UP

(C) Exactly 22

(D) Impossible to tell

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

6 **Answer in complete sentences.**

Estimate 47% of 312. Show which strategy you used — benchmark, rounding, or compatible numbers — and give your estimate.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

You sorted a move into 'Benchmark percent' and your partner into 'Compatible numbers.' Which number did each strategy change — the percent or the whole — and why does changing that one keep the estimate close?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 4.3

Estimate the Percent of a Number

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
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LEARNING TARGET I can estimate the percent of a number using benchmark percents, rounding, and compatible numbers.

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each percent problem to the benchmark combination that estimates it.

_____ 15% of 200

_____ 75% of 80

_____ 20% of 350

_____ 11% of 400

(A) 10% + 5%: 20 + 10 = 30

(B) 50% + 25%: 40 + 20 = 60

(C) 10% doubled: 35 × 2 = 70

(D) 10% + 1%: 40 + 4 = 44

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Two students estimate 8.6% of 216. Ana uses 10% of 220 = 22. Ben uses 10% of 200 = 20. Whose estimate is closer to the exact answer, 18.576?

(A) Ana's — 22 is closer to 18.576

(B) They are equally close

(C) Ben's — 20 is 1.4 away; Ana's 22 is 3.4 away

(D) Neither is a valid estimate

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

For which problem is ESTIMATING clearly the right tool, rather than exact computation?

(A) Deciding if a $58 jacket at 25% off fits your $45 budget

(B) Programming a bank to compute interest on every account

(C) Reporting a medicine dose

(D) Scoring a state test

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A store marks a $79.95 item 32% off. Which mental estimate best checks the discount?

(A) About 3.2 dollars off

(B) About 32 dollars off, because 32% means $32

(C) About one third of 80, so about $26 off

(D) About half of 80, so $40 off

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Answer in complete sentences.**

Write your own estimation problem about percent, using a real situation from your life. Then solve it with a benchmark and say whether your estimate runs high or low.

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

Margaret has 6 GB of total phone storage, of which 4 GB is used and 3.5 GB is photos. Estimate what percent of the TOTAL storage is photos. Compare it with 87.5%, the percent of the USED storage that is photos. Why do these percentages differ?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

You sorted a move into 'Benchmark percent' and your partner into 'Compatible numbers.' Which number did each strategy change — the percent or the whole — and why does changing that one keep the estimate close?

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